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PALEOMAGNETIC SECULAR VARIATION STUDY OF LAVAS FROM THE MASSIF CENTRAL, FRANCE*

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The angular standard deviation of virtual geomagnetic poles determined from a paleomagnetic study of 31 Brunhes-age lava flows from France has been obtained. The value, referred to the axial dipole (geographic axis), is 15.2° with 95% confidence limits of 18.4° and 12.9°. This value is not significantly different from that obtained in similar studies of lavas from other midlatitude localities (Western United States and New Zealand).

1. Introduction

During historic time, both the amount of secular variation of the geomagnetic field (SV) and the magnitude of the nondipole field (whose changes cause most of the historically observed SV) have been rather irregularly distributed over the earth's surface. For example, these features are at the present time of larger magnitude in the southern hemisphere than in the north and are almost completely lacking in the central Pacific region. The past distribution of such features and their permanence are of interest in posing boundary conditions for theories of the dynamo processes in the earth's core that give rise to the geomagnetic field.

This paleosecular variation study of lava flows from the Massif Central of France is one of several undertaken to study the long-term nature of SV [1–4]. The time span selected for these studies is the Brunhes normal polarity epoch, which extends from 0.68 my ago to the present. This interval is probably long enough to ascertain the overall pattern of recent secular variation, as it is several times longer than the maximum periods (10,000 yr) suspected to be involved in SV [4–6]. Detailed sequential changes in geomagnetic field directions can rarely be determined paleomagnetically; for this reason the amount of angular dispersion measured about some assumed (or mean) field distribution is the quantity of interest for a given locality.

In a recent study of paleosecular variation in New Zealand, Cox [1] showed that angular dispersion for the Brunhes epoch in the southern hemisphere (New Zealand) was not significantly different from that in Western North America. The present study is of a similar nature, undertaken to test for a possible significant difference in angular dispersion between Western North America and Western Europe.

Several paleomagnetic studies of lavas of Brunhes age from the Massif Central and elsewhere in France have previously been made for various puposes [7–10]. (Also see Roche [7], for references to several previous studies of Brunhes, David, Godard, Grenet, Roche, and the Thelliers.) The relations of some of these studies to the current sample collections will be discussed below, but because of the generally different sampling schemes and measuring techniques employed, the older data have not been combined with the present determinations in order that uniformity be maintained in the statistical analyses for overall SV. In the course of this study, collections were also made from several pre-Brunhes-age lava flows. Although not germane to the primary purpose, the results have been included in tables and illustrations.

The majority of the sample collections of the Brunhes-age lavas were made in the Chaîn des Puys

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near Clermont-Ferrand, France; a few were taken from lavas in the valleys of the Ardèche and its tributaries near Thueyts. The pre-Brunhes age lava flows were collected in the Chaîn des Puys and in southern France southwest of Montpellier. Details of the sampling sites are given in the appendix. The Carte Géologique de la France, at a scale of 1:80 000, was used as the primary geologic basis for the collections.

2. Paleomagnetic studies

The samples used in this study were geographically oriented drill cores, 2.5 cm diameter by 10 to 20 cm long [11]. Eight cores were drilled from each lava studied, with maximum lateral separations of a few tens of meters to a few hundreds of meters so as to get as representative an RM direction as possible for the entire flow. The maximum lateral separation for each volcanic rock unit studied is given in tables 1 and 2. Specimens 2.3 cm long were cut from each core and the NRM of one such specimen from each core was measured on a spinner magnetometer [12]. One or more specimens from each lava flow were then progressively partially demagnetized in alternating fields of 25, 50, 100, 200, 400 (and occasionally 600 and 800) oersteds peak field strength using equipment described by Doell and Cox [13]. Peak field strength for magnetic cleaning of the remaining samples from a given lava were chosen on the basis of the scatter in NRM directions and the behavior during partial demagnetization.

Magnetic characteristics varied considerably from lava to lava. For many flows, the NRM directions were tightly grouped and did not change after partial demagnetization. Moreover, the shapes of the demagnetization curves were characteristic of thermoremanent magnetization (TRM) [14]. In other flows, magnetic cleaning at peak field strengths up to 200 oersteds removed anomalous magnetizations so that the directions of the original TRM could be accurately determined. In several flows, however, it was not possible to remove anomalous magnetization (probably induced by lightning [15, 16]) from all samples and recover the original TRM directions. For these flows, the specimens with unremoved anomalous magnetiza-

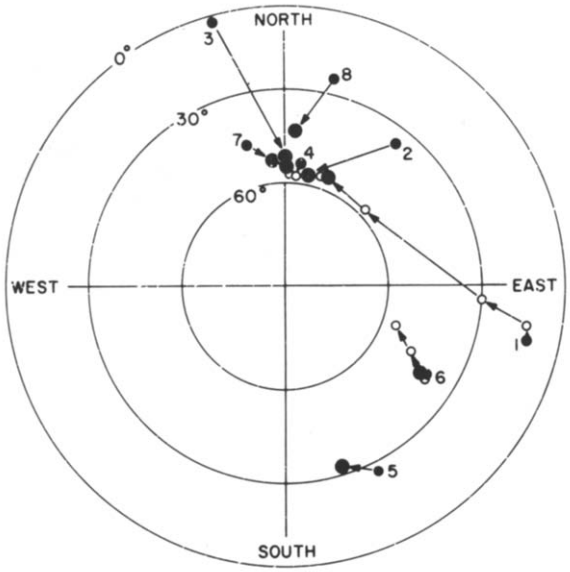


Fig. 1. Directions of remanent magnetization in samples from flow number F26 during partial alternating field (AF) demagnetization experiments. Small filled circles, natural remanent magnetization directions; large filled circles, directions of remanent magnetization after AF demagnetization at 200 peak oersteds; small open circles, directions during progressive AF demagnetization. Lower hemisphere of Lambert equal-area projection.

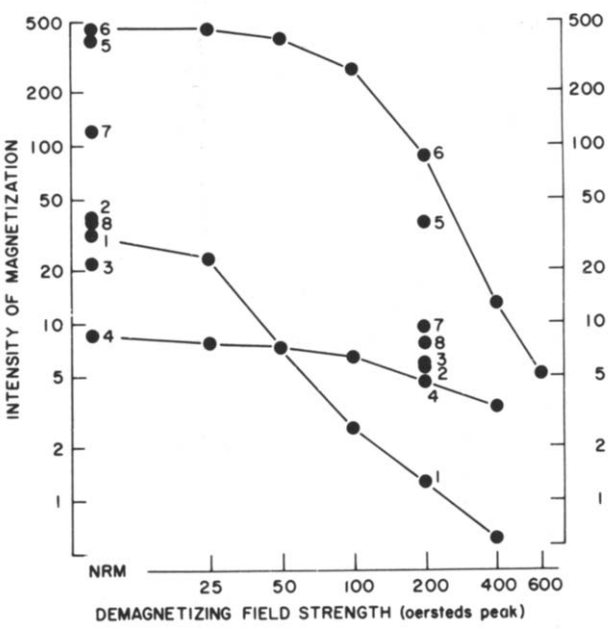


Fig. 2 Intensity of remanent magnetization of samples from flow F26 during partial AF demagnetization experiments.

Table 1
Flow-average data for Brunhes-age volcanics

Unit No.	Lateral sampling extent (m)	Number of samples used in analysis	Cleaning field strength (peak oe.)	Declination of RM (deg. E.)	Inclination of RM (deg. dn.)	Length of resultant R	Precision parameter k_c	95% confidence α_{95} (deg.)	Angular S.D. (deg.)	VGP Lat. (deg. N.)	VGP Long. (deg. E.)
F 1	58	8	NRM	358.5	55.4	7.9794	340	3.0	4.4	80.3	190.2
F 2	16	8	200	31.3	61.2	7.9394	115	5.2	7.6	67.4	90.2
F 3	64	8	50	356.8	42.4	7.9602	176	4.2	6.1	68.7	190.9
F 4	63	8	100	12.4	41.6	7.9833	419	2.7	4.0	66.0	154.2
F 5	68	8	NRM	356.3	56.0	7.9781	319	3.1	4.5	80.7	201.5
F 6	70	8	50	342.0	62.0	7.9858	494	2.5	3.6	76.9	269.4
F 7	116	8	NRM	356.7	59.8	7.9778	315	3.1	4.6	84.5	210.7
F 8	19	5	200	354.5	59.9	4.9876	323	4.3	4.5	83.8	225.1
F 9	17	8	NRM	1.5	51.8	7.9854	478	2.5	3.7	76.7	177.5
F10	38	6	50	21.6	79.3	5.9598	124	6.0	7.3	64.1	20.3
F11	70	8	100	18.0	63.0	7.9301	100	5.6	8.1	77.3	92.4
F12	45	7	50	29.3	64.8	6.9608	153	4.9	6.6	69.9	79.6
F13	58	7	200	359.0	61.3	6.9402	100	6.0	8.1	86.5	194.6
F14	49	4	200	344.6	66.9	3.9864	200	6.2	5.5	79.0	298.8
F15	61	8	50	355.5	67.6	7.9722	252	3.5	5.1	84.5	333.0
(F16*	12	6	100	356.5	57.1	5.9272	69	8.1	9.8	81.5	202.0)
F17	71	8	50	359.6	63.0	7.9755	285	3.3	4.8	88.6	193.4
F18	29	6	100	346.0	66.0	5.9771	219	4.5	5 5	80.2	292.9
F19	177	6	NRM	346.9	60.9	5.9770	217	4.6	5 5	79.8	255.2
F20	130	7	200	345.9	62.5	6.9815	325	3.4	4.5	79.8	266.5
(F21†	28	8	100	284.8	71.6	7.9518	145	4.6	6.7	44.0	314.75)
F22	73	8	200	20.1	73.7	7.9734	263	3.4	5.0	71.7	36.4
F23	40	7	50	357.1	57.1	6.9618	157	4.8	6.5	81.7	199.1
F24	50	4	200	346.6	62.3	3.9795	146	7.6	6.7	80.2	265.2
F25	66	8	NRM	14.0	60.8	7.9838	432	2.7	3.9	79.2	108.8
F26	45	6	200	5.7	52.8	5.95821	120	6.2	7.4	77.0	161.1
F27	63	8	50	343.7	67.7	7.9312	102	5.5	8.0	78.2	302.5
F28	175	4	400	335.4	71.3	3.9927	410	4.5	4.0	71.7	314.9
F29	174	8	NRM	334.6	57.3	7.9684	222	3.7	5.4	69.8	263.5
F30	18	7	100	329.9	52.5	6.9457	111	5.8	7.7	64.0	258.2
F31	60	8	NRM	9.7	48.6	7.9750	280	3.3	4.8	73.1	154.0
F32	40	8	25	10.5	41.7	7.9852	473	2.6	3.7	67.6	158.4
F33	70	7	200	335.6	50.1	6.9642	168	4.7	6.3	66.4	246.8

* Same lava as F17 – not included in statistical calculations.
† Not included in statistical analyses because direction classed as "intermediate" (see text).

tion were excluded from the analyses, the results of which are given in tables 1 and 2.

The selection of specimens to be removed from the analyses is in part subjective but was made in a manner that would yield an average direction for each lava that was most probably that of the original TRM. A typical example, that for flow F26, is considered in figs. 1 and 2. The intensities of RM of the eight specimens before and after the demagnetization treatments are given in fig. 2. The NRM direction of specimen 4, which was near the majority of the RM directions of all specimens after cleaning at 200 oersteds, did not change significantly during progressive demagnetization, and the shape of its demagnetization curve (fig. 2) is typical of TRM. In contrast, the direction of RM in specimen 1 changes rapidly towards the final "preferred" direction during partial demagnetization up to 200 oersteds peak value, but

Table 2
Flow-average data for pre-Brunhes-age volcanics.

Unit no.	Lateral sampling extent (m)	Number of samples used in analysis	Cleaning field strength (peak oe.)	Inclination of RM (deg. E.)	Declination of RM (deg. dn.)	Length of resultant R
F34	45	8	200	189.0	−46.0	7.9422
F35	30	6	200	186.5	−29.4	5.9699
F36	14	8	NRM	342.1	57.2	7.9773
F37	110	7	200	207.0	−56.5	6.9548
F38	160	5	100	358.9	67.2	4.9754
F39	70	8	200	356.8	58.8	7.9631
F40	50	8	50	155.9	−16.4	7.9329
F41	98	8	50	196.9	−68.0	7.9446
F42	40	7	200	160.6	−41.6	6.9714
F43	40	8	50	183.6	−40.5	7.9328

Table 2 (cont.)

Unit no.	Precision parameter k_c	95% confidence γ_{95} (deg.)	Angular S.D. (deg.)	VGP Lat. (deg. N.)	VGP Long. (deg. E.)	Polarity
F34	121	5.0	7.4	72.5	156.0	R
F35	166	5.2	6.3	61.7	170.0	R
F36	308	3.2	4.6	75.4	257.1	N
F37	133	5.3	7.0	68.2	106.2	R
F38	163	6.0	6.4	85.4	354.0	N
F39	190	4.0	5.9	83.8	206.1	N
F40	104	5.4	7.9	47.4	219.9	R
F41	126	5.0	7.2	77.5	61.0	R
F42	209	4.2	5.6	63.4	225.4	R
F43	104	5.4	8.0	67.2	174.5	R

does not change further with treatment at 400 oersteds peak field strength. The intensity curve falls off rapidly, suggesting removal of a large amount of relatively low coercivity anomalous magnetization, and thus it was possible to recover the original TRM direction in this specimen. The behavior of specimen 6, which had an NRM intensity about one order of magnitude higher than most specimens, suggests that its anomalous magnetization has much higher coercivity than that of specimen 1. Although the direction of the RM after demagnetization at 600 oersteds peak has moved toward the preferred direction, there is still present a large anomalous component. Judging by the direction and intensity of NRM and RM after 200 oersteds AF cleaning, it seems probable that specimens 2, 3, 7, and 8 possess characteristics between those of specimens 1 and 4, and that specimen 5 is very similar to specimen 6. It is concluded that the best estimate of the original TRM direction can be obtained by removing specimens 5 and 6 from the flow-average analysis, and this has been done for the results given in table 1. Similar reasoning led to the rejection of specimens where the "Number of specimens used in analysis" entry in tables 1 and 2 is less than eight. In two lava flows sampled, F44 and F45 (sect. 4), no original TRM direction could be estimated.

We take the flow-average directions of RM magnetization cited in tables 1 and 2 as the best estimates of the ancient field directions at the time and place

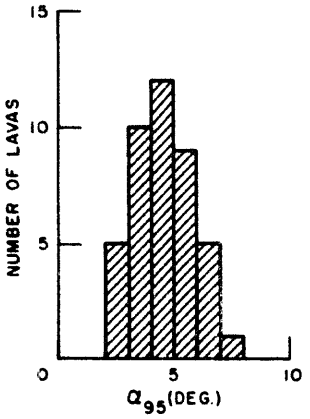


Fig. 3. Histogram of 95% confidence cone semiangles (α_{95}).

of cooling of the lavas. The statistical parameters cited, for example the semiradius of the 95% confidence cone about the mean direction, α_{95} [17], may be taken as measures of how accurately these directions have been determined. The overall precision of the data is expressed as a histogram of the α_{95} values (fig. 3). The majority of the directions are defined within a cone of some 5–6° radius (at the 95% confidence level), which is relatively small with respect to the average angles between flow mean directions. As has been noted previously [14], the directions cited in tables 1 and 2 represent only the ancient field direction averaged over the part of the flow sampled and not the regional geomagnetic field direction. The local anomalies that cause these differences between local and regional directions can no longer be determined for each flow. In the study cited above [14], however, the effect was calculated to average about $1\frac{1}{4}$°; it should not be greatly different here and will not add a great deal to the uncertainties cited. Therefore, as will be discussed more fully below, these data should be of adequate precision to estimate ancient secular variation. Virtual geomagnetic pole (VGP) positions for the lavas studied are shown in figs. 4 (Brunhes age lavas) and 5 (pre-Brunhes age lavas).

It is of interest to compare a few of the results of earlier paleomagnetic studies with those from this study. The Puy de Dôme was last studied by Roche [7], who reported a direction 33°E, 71° down. This direction differs by only 4.7° from that cited here (F22). Another lava of interest, because of an unusual direction, is the upper of two lavas east of Puy

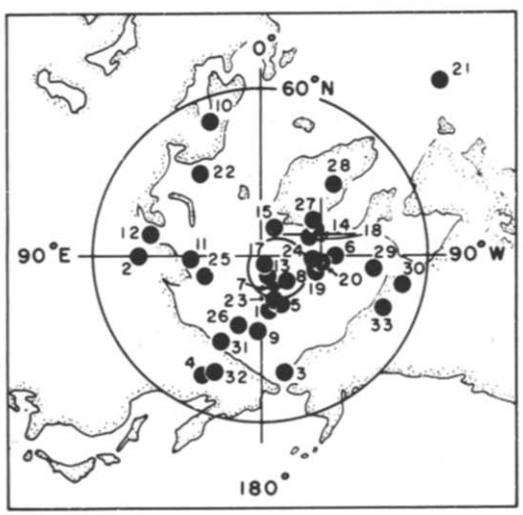


Fig. 4. Virtual geomagnetic pole (VGP) positions for Brunhes-age lavas. Lambert equal-area projection of a portion of the northern hemisphere. Large cross, position of present geomagnetic pole; open circle, 95% confidence circle about mean of all VGP's. Numbers refer to flow numbers in table 1.

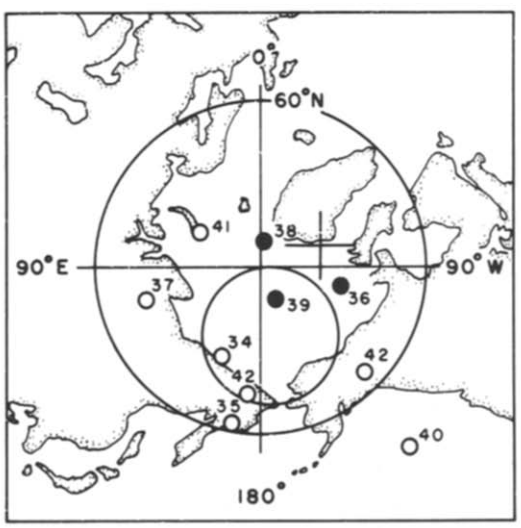


Fig. 5. Virtual geomagnetic pole positions for pre-Brunhes-age lavas. Filled circles, normal polarity; open circles, reversed polarity. See also caption for fig. 4.

Table 3
Statistical analyses of flow-average data.

Data analyzed	Latitude	Longitude	α_{95}	N	R	K_T	S_T
Brunhes-age VGP's	86.4°	212.2°	5.0°	31	29.925	28	15.4°
Pre-Brunhes-age VGP's	77.5	187.5	12.0	10	9.475	17	19.7
	Inclination	Declination	α_{95}	N	R	k_T	s_T
Brunhes-age RM directions	60.3°	357.8°	3.8°	31	30.362	47	11.9°
Pre-Brunhes-age RM directions	49.4	357.9	12.1	10	9.469	17	19.9

Notes:
α_{95} = 95% confidence angle about means
N = number of lavas
R = vector sum of N unit vectors
K_T, k_T = total precision parameter, $(N\text{-}1)/(N\text{-}R)$, virtual geomagnetic poles and directions respectively
S_T, s_T = total angular standard deviation, $(N\text{-}1)^{-\frac{1}{2}}\left(\sum_{i=1}^{N}\delta_i^2\right)^{\frac{1}{2}}$ where δ_i is the angle between the VGP and direction at site i, respectively, and the means.

de Dôme. Bonhommet and Babkin [9] reported a direction 65°W, 65° down, whereas the direction reported here (F21) is 75.2°W, 71.6° down – a difference of 7.6°p. As confidence limits have not been reported for the previous studies, significance tests cannot be made; but these differences are not large.

3. Discussion

The general paleomagnetic results from this study are summarized in table 3. VGP's and RM directions have been analyzed for both Brunhes-age lavas and pre-Brunhes-age lavas using statistical techniques common in paleomagnetic studies. The 95% confidence limit for the Brunhes-age lavas includes the geographic axis; that for the older rocks just excludes it.

Of more interest to the present study is the angular dispersion characterized by the values of the precision parameters and angular standard deviations cited in table 3. These values are the total observed variation and thus include variation due to experimental errors and that due to ancient local anomalies, as mentioned above. The total angular deviations (S_T or s_T) in this table and in table 4 were calculated using the expression

$$(N-1)^{-\frac{1}{2}}\left(\sum_{i=1}^{N}\delta_i^2\right)^{\frac{1}{2}},$$

where δ_i is the angle between the ith VGP or field direction and the position or direction about which the dispersion is desired (means or axial dipole). The total precision parameters K_T and k_T in tables 3 and 4 were calculated from the expression $(N-1)/(N-R)$ for analyses about means and the expressions $N/(N-R\cos\delta)$ for analyses about axial dipole parameters, where δ is the angle between the mean poles or directions and the geographic pole or axial dipole field direction [17]. These total dispersions are further analyzed in table 4, where the effects of experimental errors and local anomalies are removed. Experimental errors (which here include dispersions caused by intraflow anomalies) were removed by calculating the between-site dispersions from $s_B^2 = s_T^2 - s_W^2/\bar{N}$, where s_W is the mean of the individual flow angular standard deviations given in tables 1 and 2 and $\bar{N}$ the average number of samples per lava. The angular dispersion of the ancient field directions were obtained by further removing the effects of local anomalies, expressed as an angular dispersion s_A, using the relation $s_F^2 = s_B^2 - s_A^2$. S_B and S_F values for virtual geomagnetic poles were obtained

Table 4
Dispersion statistics of flow-average data.

	Angular standard deviations				Precision parameters			
Brunhes-age lavas ($\bar{N}$ = 7.06)	S_T	S_B	S_F	S_W	K_T	K_B	K_F	K_W
Referred to mean VGP	15.4°	14.9°	14.8°	10.7°	27.9	30.4	30.8	55.1
Referred to geographic pole	15.8	15.3	15.2		27.3	29.4	29.8	
	s_T	s_B	s_F	s_W	k_T	k_B	k_F	k_W
Referred to mean direction	11.9°	11.7°	11.6°	5.6°	47.0	47.8	48.4	199.3
Referred to axial dipole direction	12.3	12.2	12.1		44.1	45.5	46.0	
Pre-Brunhes-age lavas ($\bar{N}$ = 7.28)	S_T	S_B	S_F	S_W	K_T	K_B	K_F	K_W
Referred to mean VGP	19.7°	19.2°	19.1°	11.8°	17.1	18.2	18.4	48.4
Referred to geographic pole	23.7	23.3	23.2		13.3	13.8	13.9	
	s_T	s_B	s_F	s_W	k_T	k_B	k_F	k_W
Referred to mean direction	19.9°	19.7°	19.7°	6.6°	16.9	17.3	17.4	144.6
Referred to axial dipole direction	24.8	24.6	24.6		12.0	12.1	12.1	

Notes:
S, s = angular standard deviations for VGP's and directions, respectively
K, k = precision parameters for VGP's and directions, respectively
Subscript T = total dispersion
Subscript B = between flow dispersion (experimental errors only removed)
Subscript F = dispersion of VGP's and directions (experimental errors and effects of local anomalies removed)
Subscript W = within flow dispersion
$\bar{N}$ = mean number of sample cores per lava

in a similar manner. Lacking better data, we took a value of $s_A = 1.25°$ as appropriate for field directions and used methods developed by Cox [18] to transform the value of $s_A = 1.25°$ to a value $S_A = 1.77°$, which is appropriate for analysis of poles.

The between-site precision parameters K_B and k_B, when referred to means, were calculated using the two-level analysis methods given by Watson and Irving [19]; these procedures also gave the within-site precision parameters K_W and k_W. When referred to the axial dipole field direction, k_B was calculated using the expression $(1/k_B) = (1/k_T) - (1/\bar{N}\,k_W)$; a similar expression was used to calculate values of K_B for dispersion about the geographic pole. Finally, expressions of the type $(1/k_F) = (1/k_B) - (1/k_A)$ were used to remove the effects of local anomalies. Again, Cox's methods [18] were used to transform a value $k_A = 4200$ for analysis of directions to a value $K_A = 2100$ for analysis of poles. One may also relate precision parameters to angular deviations directly, using the expression $s^2 \cong 2/k$, which is valid for small dispersions. For example, the precision parameter for Brunhes-age VGP's about their mean gives an angular dispersion of $(2/30.8)^{\frac{1}{2}} = 14.6°$, which is very close to that obtained from direct analysis of the angular deviations, 14.8°. Thus, the final results are not particularly sensitive to the methods used to remove the effects of experimental errors or local anomalies, which, in fact, are relatively small compared with the angular deviations between the flow means.

In comparing paleosecular variation of one region with another, the question arises as to which measures of angular dispersion are most appropriate – precision parameters or angular deviations, directions or virtual geomagnetic poles, analyses about means or an axial dipole. There is little choice between angular dispersions or precision parameters, as both are valid

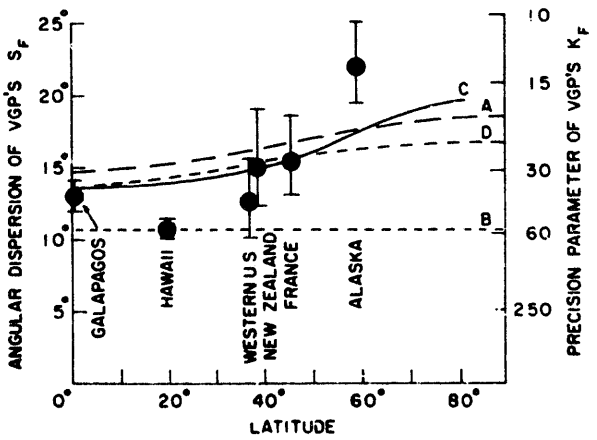


Fig. 6. Observed paleoangular standard deviation (S_F) with 95% confidence limits [23] for several localities plotted against latitude. The equivalent precision parameter scale (K_F) is given on the right. The curves labeled A, B, C, and D are expected values of S_F for several models and are discussed in the text.

measures of dispersion. The former are preferred here, however, because they vary more regularly with the impression of scatter of poles or directions when viewed in illustrations such as figs. 4 or 5.

Whether poles or directions are analyzed is perhaps of more significance. Angular dispersion of the geomagnetic field arises from two causes: the strength and direction of the nondipole field with respect to the dipole field and the wobble of the dipole itself. Dispersion of VGP's produced by dipole wobble is invariant with respect to latitude [5, 18], whereas dispersion of directions is not. On the other hand, randomly directed nondipole fields of uniform size at all latitudes, added to a fixed dipole field, produce dispersion that varies with latitude whether poles or directions are analyzed [5, 20]. Therefore, when comparing angular dispersion from different regions, it seems preferable to use analyses of VGP's rather than directions because effects of dipole wobble are the same at all localities. For example, the lowest observed VGP angular variation at any locality is a maximum value of the contribution due to dipole wobble for the entire earth. Angular variation in excess of this amount at any other locality is then a minimum value for dispersion produced by the nondipole field at that locality.

The analyses should, of course, be made about the true average mean pole for the period of time under consideration, but this is not known for paleosecular variation studies. To date, most paleomagnetic studies of Brunhes-age that probably cover a period of time sufficiently long to average out secular variation yield average poles not significantly different from the rotation axis [21, 22]; for this reason it seems preferable to analyze dispersion at all localities with respect to the geographic axis. In fig. 6 the angular deviations of flow average VGP's about the geographic pole for several localities discussed in earlier works [1, 4] are plotted against latitude. The 95% confidence limits shown as error bars were obtained following the method given by Cox [23]. These values are somewhat different from those previously cited for three reasons: (1) more data have been added in some instances; (2) all data with flow-average values of $\alpha_{95} > 9°$ have been removed in order to maintain relative statistical uniformity within individual data; (3) VGP'S with colatitudes greater than 44° have been classed as "intermediate" and excluded from analysis. These later criteria will be discussed in more detail in future works; for the present, we note only that the major effect has been to lower somewhat most of the values of dispersion previously reported. The two determinations previously available for midlatitudes are from New Zealand [1] and the Western United States [4]. The present value from France is nearly identical with that from New Zealand and not significantly different from that for the Western United States. (The ratio of the angular deviations for France and the Western United States is 1.22, whereas the appropriate F-test value is $F_{0.05}$ (60, 38) = 1.66.) Also plotted in fig. 6 are curves of four models for angular deviation versus latitude. Curve A [24] results from adding a nondipole field of amplitude invariant with latitude to a fixed axial dipole; curve B [25] is generated by dipole wobble only (and this has a maximum value equal to the lowest one observed – that for Hawaii); Curve C [5] is constructed by combining a dipole wobble of 11° and a nondipole field distribution similar to the present one; and curve D [18] adds an 11° dipole wobble to a nondipole field with the same average strength at all latitudes. The angular standard deviation recorded in the Brunhes-age lavas of the Massif Central is not significantly different from models A, C, or D, but differs considerably from model B at the 95% con-

fidence level, as do most of the other observed values in fig. 6. If the value for Hawaii (S_F = 10.8°) represents dispersion due to dipole wobble only (S_D), as has been suggested previously [2, 3], then the contribution by the nondipole field (S_N) to the total dispersion, (S_F) in France, is given by

$$S_N = \sqrt{S_F^2 - S_D^2} = 10.7^\circ ,$$

where, from table 4, S_F = 15.2°. It seems, therefore, that in midlatitudes during the Brunhes epoch, the changing nondipole field and dipole wobble contribute equally to the angular dispersion of VGP's.

Values of angular deviation for the pre-Brunhes lavas are considerably larger. But because of the small number of lavas studied and the probable wide age range covered, probably no significance should be made of this fact.

4. Appendix: Location of sampling sites

The maps referred to are from the Map of France at 1:50 000. The sheets are:

Agde	Flle	XXVI-45
Besse		XXV-33
Bourg-Lastic		XXIV-32
Burzet		XXVIII-37
Clermont-Fd.		XXV-31
Latour		XXIV-33
Pézanas		XXVI-44
Pontgibaud		XXIV-31
Privas		XXIX-37
Veyre-Monton		XXV-32

The numbers in parentheses are the latitude and longitude of the sample sites.

F1 Basalt flow in valley of la Veyre R. samples in roadcuts of r D213 1 km E of Ponteix, north center Veyre-Monton Flle. (45.638°N, 3.038°E).

F2 Basalt flow S. and E. of Puy de la Rodde. Samples in roadcuts of r D5 E. of Puy de la Rodde, N.W. Veyre-Monton Flle. (45.638°N, 2.966°E).

F3 Trachyandesite flow E. of Puy Moy. Samples in roadcuts of r N89 2 km W.-S.W. of Theix, N. center Veyre-Monton Flle. (45.699° N, 3.000° E).

F4 Basalt flow in valley of Auxon R. Sampled in roadcuts of r N89 just W. of Theix, N. center Veyre-Monton Flle. (45.700°N, 3.018°E).

F5 Basalt flow in valley of la Gaxelle R. Sampled in roadcuts of r D26 1 km S. of Valbeleix, W. center Besse Flle. (45.450°N, 2.982°E).

F6 Basalt flow in valley of Chassagne R. Sampled in roadcuts of r D23 1 km S. of Tourzel – Ronzières, N. center Besse Flle. (45.520°N, 3.126°E).

F7 Basalt flow in valley of la Couse R. Sampled in roadcuts of r D26 100 m W. of r D26E, N. center Besse Flle. (45.533°N, 3.042°E).

F8 Basalt flow S. and W. of Montchat Cratère. Sampled in natural outcrops at extreme W. end of flow near r N678, N.E. Latour Flle. (45.458°N, 2.690°E).

F9 Andesite flow S.E. of Puy de la Vache. Sampled in roadcuts of r N89 1 km W. of la Cassière, N.W.Veyre-Monton Flle. (45.693°N, 2.990°E).

F10 Trachybasalt S. of Ruines de Prada. Sampled in roadcuts of r N89 just S.E. of Nébouzat, N.W. Veyre-Monton Flle. (45.713°N, 2.909°E).

F11 Trachyandesite W. of Puy de Barme. Sampled in roadcuts of r N141A 0.5 km E. of les Quatres Routes, S.W. Clermont-Fd. Flle. (45.728°N, 2.900°E).

F12 Basalt flow W. of Puy de Barme. Sampled in roadcuts of r N141A 0.5 km E. of Bramaud, D.W.Clermont-Fd. Flle. (45.730°N, 2.915°E).

F13 Trachybasalt flow S.E. of Puy de Dôme. Sampled in natural outcrops just W. of junction of road to Puy de Dôme and r D68, S.W. Clermont-Fd. Flle. (45.768°N, 2.988°E).

F14 Trachyandesite flow E.-S.E. of Puy de Pariou. Samples from artificial cuts just N. of r N141B 1 km W. of Orcines, center Clermont-Fd. Flle. (45.790°N, 3.049°E).

F15 Trachyandesite N.E. of Puy de Pariou. Sampled in abandoned quarry on r N141B 3 km N.W. of Orcines, center Clermont-Fd. Flle. (45.800°N, 2.991°E).

F16 Trachyandesite flow N.W. of Puy de Côme. Sampled in roadcuts of r D52E 1.5 km S.W. of junction with r N141B, W. center Clermont-Fd. Flle. (45.803°N, 2.938°E). Samples not used in study (same as F17).

F17 Trachyandesite flow N.W. of Puy de Côme. Sampled in roadcuts of r D62 1 km N.E. of Banière, E. center Pontigibaud Flle. (45.811°N, 2.871°E).

F18 Basalt flow of Puy de Chanal. Sampled in roadcuts of r D90 just S. of Chanat, center Clermont-Fd. Flle. (45.827°N, 3.152°E).

F19 Basalt flow E. of Volvic. Sampled in roadcuts of r D83 1.5. km E. of Volvic, N. center Clermont-Fd. Flle. (45.872°N, 3.058°E).

F20 Trachyandesite flow E. of Puy de Louchadière. Sampled in roadcuts of r N141 at intersection of r D2 Argnat, N. center Clermont-Fd. Flle. (45.844°N, 3.024°E).

F21 Trachybasalt flow from N.E. Puy de Dôme. Sampled in roadcuts of r D58 2 km W. of Royat, S. center Clermont-Fd. Flle. (45.768°N, 3.027°E).

F22 Domite of Puy de Dôme. Sampled in roadcuts of summit road 100 m E.-N.E. of summit, S. center Clermont-Fd. Flle. (45.773°N, 2.967°E).

F23 Trachybasalt flow W. of Puy Besace. Sampled in natural exposures 50 m N. of r D52 3 km E of Ceyssat, S.W. Clermont-Fd. Flle. (45.752°N, 2.937°E).

F24 Trachybasalt flow S. of Clermont-Fd. Sampled in an abandoned quarry on r N9 3 km S.E. of Clermont-Fd., S. center Clermont-Fd. Flle. (45.760°N, 3.121°E).

F25 Basalt flow in valley of la Couxe de Chambon R. Sampled in roadcuts of r N496 0.5 km E. of Verrières, S. center Veyre-Monton Flle. (45.575°N, 3.043°E).

F26 Basalt N. of Puy de Sevière. Sampled in roadcuts of r N683 at sharp bend N.W. of Puy de Servière, N.E. Bourg-Lastic Flle. (45.653°N, 2.857°E).

F27 Trachyandesite N. of Puy de la Nugere. Sampled in roadcuts of r D92 1 km of junction with r. N686, N. center Clermont-Fd. Flle. (45.868°N, 2.995°E).

F28 Trachyandesite E. of Puy de la Nugere. Sampled in roadcuts of r D92 200 m N. of junction with r N686, N. center Clermont-Fd. Flle. (45.861°N, 3.005°E).

F29 Basalt flow N.E. of Coupe d'Aizac. Sampled in roadcuts of r D254 500 m N.W. of le Pont de l'Huile, W. center Privas Flle. (44.720°N, 4.349°E).

F30 Basalt flow in valley of la Volane R. Sampled in roadcuts of unnumbered road crossing la Volane R. 1 km S. of Raccource, 400 m S. of bridge, W. center Privas Flle. (44.699°N, 4.372°E).

F31 Basalt flow in valley of le Lignon R. Sampled in natural outcrops along river bank W. of r D5 1 km S. of junction of le Lignon R. with Ardèche R., S.E. Burzet Flle. (44.658°N, 4.271°E).

F32 Basalt flow in valley of Ardèche R. Sampled in natural outcrops along river bank just S.E. of junction of r D26 with r N102, S.E. Burzet Flle. (44.672°N, 4.252°E).

F33 Basalt flow in valley of Fontolière R. Sampled in roadcuts of r N536 400 m E. of junction of r D26, S.E. Burzet Flle. (44 690°N, 4 265°E).

F34 Basalt flow S.E. of Mt. St. Loup. Sampled in natural outcrops in the surf at Cap d'Agde, E. center Agde Flle. (43.275°N, 3.516°E).

F35 Basalt N. of Valros. Sampled in an abandoned quarry 200 m E. of r NP9 1 km N. of Valros, S. center Pézenas Flle. (43.429°N, 3.433°E).

F36 Basalt flow south of St. Thibéry. Sampled in quarry 300 m W. of r D137 and 300 m S. of Naffrie, S. center Pézenas Flle. (43.382°N, 3.433°E).

F37 Basalt flow E. of Zanière. Sampled in roadcuts of r D23 3 km N.W. of Chausse, center Besse Flle. (45.432°N, 3.098°E).

F38 Basalt flow E. of Rentières. Sampled in roadcuts of r D23 200 m S.E. of Chausse, center Besse Flle. (45.413°N, 3.128°E).

F39 Basalt E. of St.Alyre. Sampled in roadcuts of r D36 E. of l'Argillier, S. center Besse Flle. (45.395°N, 3.018°E).

F40 Basalt E. of Courgoul. Sampled in roadcuts of r D28 100 m E. of junction with r D28E, N. center Besse Flle. (45.519°N, 3.147°E).

F41 Ankaramite W. of St. Diéry. Sampled in roadcuts of r D5E 1 km S.W. of junction with r D146, S.W. Veyre-Monton Flle. (45.548°N, 2.989°E).

F42 Basalt W. of Picherande. Sampled in small quarry at edge of r D203 1 km W. of Picherande, N.E. Latour Flle. (45.465°N, 2.761°E).

F43 Basalt S.E. of Ternant. Sampled in an abandoned quarry on r D90 0.5 km S.E. of Ternant, center Clermond-Fd. Flle. (45.804°N, 3.152°E).

F44 Trachyandesite flow S.W. of Puy de Côme. Sampled in natural outcrops near r D52 center Clermont-Fd. Flle. (45.781°N, 2.879°E). No stable RM component could be identified in these samples; they therefore have not been used in this study.

F45 Trachybasalt flow N.E. of Ceyssat. Sampled in natural exposures 0.5 km N. of r D52, 1 km E. of Ceyssat, S.W. Clermont-Fd. Flle. (45.770°N, 2.908°E). No stable RM component could be identified in these samples; they therefore have not been used in this study.

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